Where quantum amplitude estimation meets a PSA decision

Matched estimands expose estimator performance, reversible-computing cost, and validation limits

1 Frozen PSA decision

Recycle policies: 0.420 / 0.427 / 0.430
Joint event: H2 purity + recovery
Correlated process uncertainty
Chance target: p >= 0.50

2 Matched probability estimators

200 replications per method

MC	RMSE 0.00679
RQMC	RMSE 0.00248 | best
IS	RMSE 0.01017
MLAE	RMSE 0.00289 | ideal

3 Measured implementation boundary

Response oracle: 99 q / 35,160 CX
Parameter-only CSS: 32/32 events
5-component LDF modules: 675 q / 6.55 M CX
Valve→boundary: 842 q / 2.35 M CX | tank: 677 q
Gas→y→2 faces→cell: 1,651 q / 14.67 M CX
Direct LDF→cell balance: 1,317 q / 19.08 M CX
Policy stream→event: 510 q / 1.29 M CX
Raw agg. CSS: 2/32 | physical: 68→85 cycles
Out-of-place state/streams: 2,880 q
Dense 7-bit loader: 2^63 amplitudes
IBM 3x3: mean 0.59919 vs 0.625

Current result: no end-to-end quantum advantage
State accumulation/full phase logic, structured loading, and H2 validation remain decisive